

Hybrid Mechanistic & Machine-Learning Optimization System for Pectin Extraction

Virtual Reactor (V1.3 Digital Twin) Mathematical Specification

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1. Initial & Boundary Conditions ($t = 0$)

The initial state establishes the mathematical invariants for the fresh rind mass balance.

- $X_{\{w,0\}}$: Initial moisture content (wet basis) $\approx 0.81 - 0.86$
- $P_{\{\text{matrix},0\}}$: Initial available protopectin content $[\text{kg}/\text{kg}_{\{\text{dry,feed}\}}]$
- $X_{\{\text{GalA,matrix}\}}$: GalA fraction of the matrix pectin
- $DE_{\{\text{matrix}\}}$: Degree of esterification of the matrix pectin

Explicit Initial Conditions (Masses):

- $P_{\{\text{matrix}\}}(0) = P_{\{\text{matrix},0\}}$
- $P_{\{\text{sol}\}}(0) = P_{\{\text{lowMW}\}}(0) = P_{\{\text{loss}\}}(0) = 0$
- $G_{\{\text{matrix}\}}(0) = P_{\{\text{matrix},0\}} \cdot X_{\{\text{GalA,matrix}\}}$
- $G_{\{\text{sol}\}}(0) = G_{\{\text{lowMW}\}}(0) = G_{\{\text{loss}\}}(0) = 0$
- $Q_{\{DE,\text{sol}\}}(0) = 0$
- $M_{\{w,\text{sol}\}}(0) = M_{\{w,\text{matrix}\}}$

Note: Literature reports for extracted passion-fruit pectin ($DE \approx 74.5\%$, $M_w \approx 6.54 \times 10^5$ Da, GalA $\approx 78\%$) are Informative Priors for the extracted-product properties, not assumptions about the fresh rind polymer. DE and MW for the soluble pool are properties of the material entering the pool and are undefined until $P_{\{\text{sol}\}} > 0$.

2. Feedstock Conditioning Submodel

2.1. Drying

Moisture removal is tracked explicitly:

$$\frac{dX_w}{dt} = -k_d(T_{\{\text{air}\}}, v_{\{\text{air}\}}) \cdot (X_w - X_{\{w, \text{eq}\}})$$

2.2. Milling

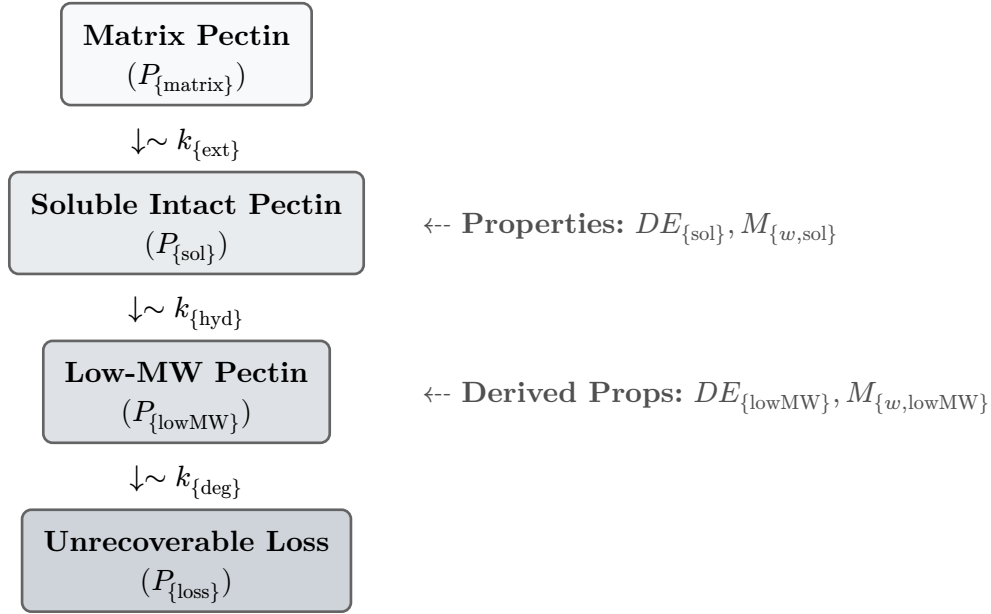
Milling reduces the bulk dried rind to a characteristic particle size ($d_{\{50\}}$). The model uses a fixed reference size for V1:

$$d_{\{\text{ref}\}} = 300 \text{ } \mu\text{m}$$

3. Acid Extraction Submodel

3.1. Conceptual Architecture

The physical extraction is modeled as a 4-pool mass transfer system, distinguishing strictly between **mass inventories** (states) and **chemical attributes** (properties).



Terminology:

- **Hydrolysis** (k_{hyd}) : Continuous acid-catalyzed chain scission within the soluble pectin population. Material crossing the model's threshold is transferred to the low-MW lumped compartment.
- **Degradation** (k_{deg}) : Low-MW pectin $\rightarrow$ non-pectic soluble products.

3.2. Differential Equations (Normalized Mass Conservation)

All P_i variables use a consistent normalized mass convention $[\text{kg}/\text{kg}_{\text{dry,feed}}]$.

1. Solubilization:

$$\frac{dP_{\{\text{matrix}\}}}{dt} = -k_{\{\text{ext}\}}P_{\{\text{matrix}\}}$$

2. Hydrolysis:

$$\frac{dP_{\{\text{sol}\}}}{dt} = k_{\{\text{ext}\}}P_{\{\text{matrix}\}} - k_{\{\text{hyd}\}}P_{\{\text{sol}\}}$$

3. Depolymerization:

$$\frac{dP_{\{\text{lowMW}\}}}{dt} = k_{\{\text{hyd}\}}P_{\{\text{sol}\}} - k_{\{\text{deg}\}}P_{\{\text{lowMW}\}}$$

4. Unrecoverable Loss:

$$\frac{dP_{\{\text{loss}\}}}{dt} = k_{\{\text{deg}\}}P_{\{\text{lowMW}\}}$$

Invariants & Diagnostics:

- **Analytical Identity:**

$$\frac{dP_{\{\text{tot}\}}}{dt} = 0$$

where $P_{\{\text{tot}\}} = P_{\{\text{matrix}\}} + P_{\{\text{sol}\}} + P_{\{\text{lowMW}\}} + P_{\{\text{loss}\}}$. *The total modeled pectin-derived mass accounting inventory is strictly conserved.*

- **Conservation Check:**

$$\varepsilon_M = P_{\{\text{matrix},0\}} - P_{\{\text{tot}\}} \approx 0$$

- **Positivity & Bounds:** $P_{i(t)} \geq 0$, $0 \leq DE_i \leq 1$, $0 \leq X_{\{\text{GalA},i\}} \leq 1$, $M_w \geq M_{\{w,\min\}} > 0$.

3.3. Cohort-Correct Properties (Moments)

Because pectin continuously enters the soluble pool, material has experienced different cumulative exposures. We model properties using mass-weighted moments.

- **De-esterification Moments:**

$$\frac{dQ_{\{DE,\text{sol}\}}}{dt} = k_{\{\text{ext}\}}P_{\{\text{matrix}\}}DE_{\{\text{matrix}\}} - k_{\{\text{de}\}}Q_{\{DE,\text{sol}\}} - k_{\{\text{hyd}\}}Q_{\{DE,\text{sol}\}}$$

$$\frac{dQ_{\{DE,\text{lowMW}\}}}{dt} = k_{\{\text{hyd}\}}Q_{\{DE,\text{sol}\}} - k_{\{\text{de}\}}Q_{\{DE,\text{lowMW}\}} - k_{\{\text{deg}\}}Q_{\{DE,\text{lowMW}\}}$$

$$DE_{\{\text{sol}\}} = \frac{Q_{\{DE,\text{sol}\}}}{P_{\{\text{sol}\}}}$$

,

$$DE_{\{\text{lowMW}\}} = \frac{Q_{\{DE,\text{lowMW}\}}}{P_{\{\text{lowMW}\}}}$$

Note: V1 assumes the de-esterification rate constant is identical for soluble intact and low-MW pectin.

- **Molecular Weight Moments:**

$$\frac{dQ_{\{MW,\text{sol}\}}}{dt} = k_{\{\text{ext}\}}P_{\{\text{matrix}\}}M_{\{w,\text{matrix}\}} - k_{\{\text{hyd}\}}Q_{\{MW,\text{sol}\}} - \beta_{\{MW\}}k_{\{\text{hyd}\}}(Q_{\{MW,\text{sol}\}} - M_{\{w,\min\}}P_{\{\text{sol}\}})$$

$$\frac{dQ_{\{MW, \text{lowMW}\}}}{dt} = k_{\{\text{hyd}\}} Q_{\{MW, \text{sol}\}} - k_{\{\text{deg}\}} Q_{\{MW, \text{lowMW}\}} - \beta_{\{MW\}} k_{\{\text{deg}\}} (Q_{\{MW, \text{lowMW}\}} - M_{\{w, \text{min}\}} P_{\{\text{lowMW}\}})$$

$$M_{\{w, \text{sol}\}} = \frac{Q_{\{MW, \text{sol}\}}}{P_{\{\text{sol}\}}}$$

,

$$M_{\{w, \text{lowMW}\}} = \frac{Q_{\{MW, \text{lowMW}\}}}{P_{\{\text{lowMW}\}}}$$

Note: This moment formulation elegantly avoids the $\frac{1}{P}$ singularity at $t = 0$. The $k_{\{\text{deg}\}}$ term implies degradation continues reducing MW toward $M_{\{w, \text{min}\}}$ in the low-MW transition zone.

3.4. Kinetics & Empirical Extraction Modifiers

Temperature and acidity are parameterized using a reference-state formulation to improve parameter inference numerical stability by decoupling prefactors and activation energies. The model architecture reserves an explicit citric-acid concentration effect ($C_{\{\text{citric}\}}$) on de-esterification; this effect is disabled in V1 by fixing $\gamma_{\{\text{citric}\}} = 0$.

- **Effective Extraction Rate:** $k_{\{\text{ext}\}} = k_{\{\text{ext}, \text{chem}\}} \varphi_d$
- **Empirical Transport Modifier:** $\varphi_d = \left(\frac{d_{\{\text{ref}\}}}{d_{\{50\}}} \right)$

Note: This is an empirical lumped accessibility/transport correction, not a mechanistic diffusion coefficient.

- **Reference State:** $T_{\{\text{ref}\}} = 353.15 \text{ K}$, $pH_{\{\text{ref}\}} = 2.0$
- **Reference Arrhenius Form:**

$$k = k_{\{\text{ref}\}} \exp \left[-\frac{E}{R} \left(\frac{1}{T_K} - \frac{1}{T_{\{\text{ref}\}}} \right) \right] 10^{\{n(pH_{\{\text{ref}\}} - pH)\}}$$

- **De-esterification:**

$$k_{\{\text{de}\}} = k_{\{\text{de}, \text{ref}\}} \exp \left[-\frac{E_{\{\text{de}\}}}{R} \left(\frac{1}{T_K} - \frac{1}{T_{\{\text{ref}\}}} \right) \right] 10^{\{n_{\{\text{de}\}}(pH_{\{\text{ref}\}} - pH)\}} (1 + \gamma_{\{\text{citric}\}} C_{\{\text{citric}\}})$$

4. Target Metrics (Normalized)

The downstream recovery factor ($\eta_{\{\text{recovery}\}}$) is applied at the product stage to account for precipitation/washing/drying losses.

- **Gravimetric Yield:** $Y_p = \eta_{\{\text{recovery}\}} (P_{\{\text{sol}\}} + P_{\{\text{lowMW}\}})$
- **GalA Inventories (Tracking Moments):**

$$\frac{dG_{\{\text{sol}\}}}{dt} = k_{\{\text{ext}\}} P_{\{\text{matrix}\}} X_{\{\text{GalA,matrix}\}} - k_{\{\text{hyd}\}} P_{\{\text{sol}\}} X_{\{\text{GalA,sol}\}}$$

$$\frac{dG_{\{\text{lowMW}\}}}{dt} = k_{\{\text{hyd}\}} P_{\{\text{sol}\}} X_{\{\text{GalA,sol}\}} - k_{\{\text{deg}\}} P_{\{\text{lowMW}\}} X_{\{\text{GalA,lowMW}\}}$$

$$\frac{dG_{\{\text{loss}\}}}{dt} = k_{\{\text{deg}\}} P_{\{\text{lowMW}\}} X_{\{\text{GalA,lowMW}\}}$$

$$X_{\{\text{GalA},i\}} = \frac{G_i}{P_i}$$

Note: In V1, the GalA fraction of matrix-associated pectin is constant during solubilization.

- **Purity:** $\text{Purity}_{\{\text{GalA}\}} = X_{\{\text{GalA,product}\}}$
- **Useful GalA Recovery:** $Y_{\{\text{GalA,extract}\}} = \frac{G_{\{\text{sol}\}} + G_{\{\text{lowMW}\}}}{G_{\{\text{matrix},0\}}}$ $Y_{\{\text{GalA,product}\}} = \frac{\eta_{\{\text{recovery}\}}(G_{\{\text{sol}\}} + G_{\{\text{lowMW}\}})}{G_{\{\text{matrix},0\}}}$

5. Parameter Ontology Table

To avoid mathematical ambiguity, all parameters are strictly assigned an ontology classification.

Symbol	Meaning	Units	V1 Anchor/ Range	Classification
T	Extraction Temperature	°C	45 – 100 °C	Controlled Process Variable
pH	Kinetic Acidity	pH	1.2 – 4.5	Controlled Process Variable
$C_{\{\text{citric}\}}$	Acid Concentration	mol/L	Initial design range = 0.05 – 0.5 M	Controlled Process Variable
t	Extraction Time	min	5 – 120 min	Controlled Process Variable
$X_{\{w,0\}}$	Fresh Moisture	% wb	81 – 86%	Measured
$d_{\{50\}}$	Particle Size	µm	149 – 500 µm	Measured
$d_{\{\text{ref}\}}$	Reference Particle Size	µm	300 µm	Fixed
$M_{\{w,\text{matrix}\}}$	MW of matrix pectin	Da	6.54×10^5	Informative prior / Inferred
$k_{\{\text{ext,ref}\}}, k_{\{\text{hyd,ref}\}}, k_{\{\text{deg,ref}\}}, k_{\{\text{de,ref}\}}$	Reference Rates	min ⁻¹	TBD	Calibrated

$E_{\text{ext}}, E_{\text{deg}}, E_{\text{hyd}}, E_{\text{de}}$	Activation Energies	J/mol	TBD	Calibrated
$n_{\text{ext}}, n_{\text{deg}}, n_{\text{hyd}}, n_{\text{de}}$	Acidity Orders	-	1.0	Fixed V1
α	Particle Size Scaling	-	0.5 – 2.0	Calibrated
β_{MW}	Hydrolysis-to-MW sensitivity	-	1.0	Fixed V1
γ_{citric}	Citric DE Multiplier	L/mol	0.0	Fixed V1
η_{recovery}	Recovery Efficiency	-	0.90	Measured / Fixed Protocol

Physical Constants:

$$R = 8.314462618 \text{ J/mol K}$$

$$T_K = T_C + 273.15$$

$$T_{\text{ref}} = 353.15 \text{ K}$$

$$pH_{\text{ref}} = 2.0$$